

SE Lab 1

PROBLEM STATEMENT #23 | Smart Cities, Transport & Logistics

Hyperlocal Courier Dispatch & Tracking Engine

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Use-Case Flow

Problem Statement: #23 – Hyperlocal Courier Dispatch & Tracking Engine

Use Case: Request Pickup (includes «Match Nearest Rider» ; extended by «cancel Request»)

Primary Actor: Sender Client

Supporting Actor: Delivery Rider (the rider ultimately matched to the request)

Preconditions: Sender Client is logged in and has entered a valid pickup address, drop-off address, and package details.

Main Success Scenario

1. Sender Client enters the pickup location, drop-off location, and package details, then submits a courier request.
2. System validates the pickup and drop-off addresses.
3. System invokes Match Nearest Rider («include»), searching for active riders within a 3 km radius of the pickup point.
4. System identifies the nearest available rider and sends a dispatch notification containing the pickup location.
5. Delivery Rider receives the notification and accepts the pickup.
6. System confirms the match to the Sender Client and displays the assigned rider's details and estimated arrival time.
7. System begins live tracking of the shipment for the Sender Client.
8. Use case ends successfully.

Alternate Flow — 3a No Rider Available in Radius

- a) System finds no active riders within the 3 km radius.
- b) System notifies the Sender Client that no riders are currently available within range and offers to retry the request or schedule the pickup for a later time.

Alternate Flow — 5a Rider Rejects or Times Out

- a) Delivery Rider rejects the dispatch, or does not respond within 60 seconds.
- b) System automatically reassigns the request to the next-nearest active rider within the 3 km radius and repeats from step 4.
- c) If no riders remain available after reassignment attempts, the flow continues from Alternate Flow 3a.

Extension Flow — «Cancel Request» (extends Request Pickup)

Extension Point: Any point after the request is submitted (step 1) and before the Delivery Rider accepts the pickup (step 5).

- a) Sender Client selects the option to cancel the pending courier request.
- b) System checks whether a Delivery Rider has already accepted the dispatch.
- c) If no rider has accepted yet, System cancels the request immediately and returns the Sender Client to the request screen.
- d) If a rider has already been dispatched but not yet accepted, System withdraws the dispatch notification from that rider before cancelling.

- e) System sets the request status to Cancelled and notifies the Sender Client that the cancellation succeeded.

Postcondition (extension): Request status is Cancelled; no rider remains dispatched to this request.

Postconditions (success): Request is matched to an active rider, the rider is notified, and live tracking has begun.

Postconditions (failure): No rider is assigned; the Sender Client is notified and the request remains open for retry or scheduling.